Supplementary Material for *From Quantum Shots to Training Data: Reorganizing Measurement Records in Quantum Machine Learning*

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a) Scope and artifact status.: This document records the exact algebra, shared four-task protocol, completed simulation, and provenance behind the main paper. The manuscript evidence contains 16,605 replicate rows and 17,010 aggregation-spectrum rows for Mackey–Glass, Santa Fe laser, Rössler, and sunspots. All four simulation tasks use the same six-qubit reservoir, features, budgets, chronological joint tuning, parameter grids, seed design, and test-access rule; separate source manifests preserve their data and implementation provenance. At the balanced setting, Grouped is supported on all four tasks. The completed unmitigated hardware transfer applies a separate fixed ten-qubit protocol to the same four tasks on two processors. Its block-level intervals describe shot variation within one acquisition, not independent hardware campaigns.

Appendix A
Exact Identity and Implementation Scaling

For aggregation scale k, let $G = N/k$ and let $x_t^{(g)} \in \mathbb{R}^d$ denote the leaked and lagged feature vector of group g at training time t. With an unpenalized intercept b, every time step receives total weight one:

$$\mathcal{L}_k(b, w) = \frac{1}{|\mathcal{T}|G} \sum_{t\in\mathcal{T}} \sum_{g=1}^{G} \big(y_t - b - w^\top x_t^{(g)}\big)^2 + \lambda\|w\|_2^2. \quad (1)$$

Define $\bar{x}_t = G^{-1}\sum_g x_t^{(g)}$ and

$$C_k = \frac{1}{|\mathcal{T}|G} \sum_{t,g} (x_t^{(g)} - \bar{x}_t)(x_t^{(g)} - \bar{x}_t)^\top. \quad (2)$$

a) Proof of the exact decomposition.: Write $x_t^{(g)} = \bar{x}_t + d_t^{(g)}$, where $\sum_g d_t^{(g)} = 0$. Expanding the square in (1), the cross term is proportional to $(y_t - b - w^\top\bar{x}_t)w^\top\sum_g d_t^{(g)}$ and therefore vanishes at every t; the remaining deviation term is $G^{-1}\sum_g (w^\top d_t^{(g)})^2 = w^\top C_{k,t} w$. Averaging over t gives

$$\mathcal{L}_k(b, w) = \frac{1}{|\mathcal{T}|} \sum_t (y_t - b - w^\top\bar{x}_t)^2 + w^\top C_k w + \lambda\|w\|_2^2, \quad (3)$$

which is deterministic and assumes neither Gaussian noise, isotropy, nor independence across time.

For a fixed set of N shot-level feature vectors on one common scale, define $S_{\text{shot},t} = N^{-1}\sum_n (x_t^{(n)} - \bar{x}_t)(x_t^{(n)} - \bar{x}_t)^\top$. A group mean is a sample without replacement, so the finite-population covariance formula gives

$$\mathbb{E}_{\mathcal{P}}[C_{k,t}] = \frac{N-k}{k(N-1)} S_{\text{shot},t}. \quad (4)$$

The factor is one for Raw and zero for EV. This statement precedes the representation-specific standardization used to fit each candidate k.

A. Scaling used in the implementation

For every fold and candidate k, standardization is fit on grouped training rows only. The same transformed rows define both C_k and the grouped fit. The implementation uses `sklearn.linear_model.Ridge`, whose loss is a sum rather than the mean loss in (1). It therefore passes

$$\texttt{alpha}_{\text{sklearn}} = \lambda\,|\mathcal{T}|G. \quad (5)$$

This preserves the meaning of λ as G changes. The generalized-ridge solver independently reconstructs the solution from $\bar{x}_t$, C_k, and λI; unit tests require its predictions to match direct grouped Ridge to numerical precision.

The directly tuned Cov-EV baseline is intentionally different. It fits EV rows with $\alpha w^\top\widehat{\Sigma}_{\text{shot}} w + \lambda\|w\|_2^2$ and tunes (α, λ) on the same chronological folds. It does not inherit Grouped training's k, C_k, or tuned penalty and is therefore a genuine competing baseline rather than an identity check. The decomposition is consistent with the general connection between feature noise and Tikhonov regularization [1], but here it holds for the observed finite record exactly.

Appendix B
Frozen Evaluation Protocol

The primary experiment was frozen in `configs/qai_robust_protocol.json` before inspecting redesigned forecast outcomes. All methods reuse the same sampled local-Z measurement record. The classical baseline receives the scalar input with the identical leak $\eta = 0.2$ and ten causal lags. The primary quantum representation contains six local Z_i and five neighboring Z_iZ_{i+1} features reconstructed from the same shots. Table I lists the three fixed-budget settings and their admissible aggregation scales.

TABLE I
THREE PREDECLARED SETTINGS. EVERY RECORD USES EXACTLY $TN = 10{,}000$ CIRCUIT EXECUTIONS.

Setting	T	N	candidate k
Shot rich	250	40	1,2,4,5,8,10,20,40
Balanced	500	20	1,2,4,5,10,20
Data rich	1000	10	1,2,5,10

A. Chronological boundaries and normalization

Indices in Table II are zero-based, half-open intervals. After washout $t_0 = 30$, the last 20% of the usable sequence is the untouched test block. The first half of the pre-test interval initializes an expanding training prefix; the remainder is split into three consecutive blocked development folds. Input normalization is frozen using raw series values only through the first training stop (inclusive because the next-step target is formed from the following value). One train-fitted affine min–max map is used on both branches. Circuit inputs alone are clipped to $[0, 1]$ for angle encoding; next-step targets retain the same affine scale without clipping and may therefore lie below zero or above one. Feature standardizers are then fit separately for every fold and candidate using training rows only.

For each candidate pair, development NRMSE is computed from one forecast per time step after averaging the group predictions. Mean development NRMSE across the three folds determines the tuned pair; exact ties prefer smaller λ, then smaller k (or smaller α for Cov-EV). The tuned configuration is refit on $[30, t_{\text{test}})$, and the final test block is evaluated once.

B. Grids and nested uncertainty

The shared ridge grid is $\lambda \in \{10^{-3}, 10^{-2}, 10^{-1}, 1, 10, 10^2, 10^3\}$. Cov-EV uses $\alpha \in \{0, 0.01, 0.03, 0.1, 0.3, 1, 3, 10\}$. Grouped searches all divisors of N, including $k = 1$ and $k = N$ as safe endpoint fallbacks. The compared methods are: matched classical input, ideal statevector QRC, EV, adaptive Grouped, Raw, and tuned Cov-EV.

Twenty independent trajectories are used for Mackey–Glass and five for Rössler. The two real-world datasets use five windows with disjoint test blocks: sunspot origins are separated by 240 months, and Santa Fe origins by 1,001 samples. Santa Fe uses only the 9,093-point continuation file, so its complete input windows are also disjoint at every budget. Reservoir seeds $\{1729, 2718, 31415\}$, shot seeds $\{1009, 2003, 3001\}$, and primary grouping seeds $\{17, 43, 97\}$ are nested factors. The inferential unit is the synthetic trajectory or measured-series window: nested repeats are averaged within each unit, and 95% t intervals are computed over the unit-level means. Paired method differences are formed within the same unit before interval estimation; shot or grouping seeds are never counted as independent forecasting samples. All forecasts predict the next step from the observed input at time t is available when predicting $t + 1$. We do not evaluate autonomous multistep rollouts.

APPENDIX C
EXPRESSIVE ARCHITECTURE AND TARGET-FREE DIAGNOSTICS

The redesigned reservoir was chosen using ideal feature behavior and hardware depth only, before forecast targets were examined. It uses six qubits, two data-reuploading layers, and seed 1729. Each layer applies input-dependent R_Y and R_Z rotations to every qubit, heterogeneous nearest-neighbor R_{ZZ} rotations on a line, and fixed heterogeneous local $R_X R_Y R_Z$ mixing. This is a compact data-reuploading construction [2]; no trainable quantum parameters are used. Table III records the frozen circuit identity.

The complete double-precision arrays for the two sets of six input-Y weights, input-Z weights, $2 \times 6 \times 3$ local angles, and 2×5 couplings are serialized in `results/expressive_reservoir/ideal_diagnostics.json`. Hash validation is mandatory in the simulation runner. Regenerating angles from the seed is convenient, but the serialized values and hash, rather than a library-specific random-number implementation, define the circuit.

A. Pre-forecast validity gate

The diagnostic evaluates 257 inputs using exact statevectors and uses no forecast target or shot sampling. All 11 features vary with input and all 11 leave more than 5% variance unexplained by an affine function of the input. The standardized feature matrix has numerical rank 11 and effective rank 5.15. Removing the second layer's couplings changes every measured feature; the maximum absolute effect is 1.58. Finally, after applying the same leak and ten-lag map, projection onto the full linear span of the filtered scalar input leaves relative residual 0.786. These values pass the frozen thresholds in the diagnostic JSON. They establish only that the measured representation is input-dependent, nonlinear, non-collapsed, and not a linear copy of the matched classical preprocessing. They are not a forecast result or a quantum-advantage claim.

APPENDIX D
COMPLETED SIMULATION OUTCOME

The manuscript filter retains only the expressive $Z + ZZ$ representation and the balanced setting. Diagnostic and ablation rows cannot enter the main report. Nested reservoir, shot, and grouping seeds are averaged within each trajectory or real-series window before the method means and paired Student-t intervals are formed. Table IV reports the resulting mean test errors.

Table V gives the two paired comparisons. A positive EV-minus-Grouped value favors Grouped; a positive Grouped-minus-Cov-EV value favors Cov-EV. Grouped has supported gains on all four tasks. Every lower bound exceeds zero. Santa Fe has the largest balanced gain.

The covariance-control gate passed its narrow condition: every upper bound in the last column was below 0.05. Cov-EV nevertheless had the lower mean on all four tasks and a supported advantage on Mackey–Glass and Rössler. The gate means only that its advantage did not exceed the frozen

TABLE II
EXACT FOLD BOUNDARIES GENERATED BY THE FROZEN PROTOCOL. EACH ENTRY IS $[\mathrm{start}, \mathrm{stop})$.

Setting	normalization stop	fold 1: fit / develop	fold 2: fit / develop	fold 3: fit / develop	final test
Shot rich	118	[30,118) / [118,147)	[30,147) / [147,177)	[30,177) / [177,206)	[206,250)
Balanced	218	[30,218) / [218,281)	[30,281) / [281,343)	[30,343) / [343,406)	[406,500)
Data rich	418	[30,418) / [418,547)	[30,547) / [547,677)	[30,677) / [677,806)	[806,1000)

TABLE III
FROZEN CIRCUIT IDENTITY.

Field	Frozen value
Circuit version	`expressive_reuploading_v2`
Qubits / layers	6 / 2
Primary seed	1729
Observables	$Z_0, \ldots, Z_5$ and $Z_i Z_{i+1}$, $i = 0, \ldots, 4$
Parameter hash	`fdff90afa7f488026e7c822f33e424682c2c084653d3ccd245f91e2e90038505`
Architecture hash	`d44fac7ba013f8337e293d9e08ca3b3bd85be634520d7b9f1953be296770b964`

practical margin; it does not show that Grouped is the uniquely effective regularizer.

The ideal-QRC control remains mixed. At the balanced setting, the classical-minus-ideal differences were -0.0921 for Mackey–Glass, 0.2688 for Santa Fe, 0.0420 for Rössler, and -0.0867 for sunspots. The matched classical input was significantly better on Mackey–Glass and sunspots; ideal QRC was significantly better on Santa Fe; Rössler's interval crossed zero. Feature validity therefore did not translate into uniform forecasting utility or establish quantum advantage.

APPENDIX E
GROUPING ASSIGNMENT AND ROBUSTNESS CONTRACT

A sampler returns an unordered set of bitstrings at each time step. Treating the first stored shot at time t and the first stored shot at $t+1$ as the same persistent shot index creates a classical bookkeeping link, not a physical trajectory. A fixed-group construction based on those positions could therefore make grouping depend on an arbitrary storage order.

For each aggregation scale k and grouping seed, the primary procedure independently permutes the N shot indices at every time step, divides the permuted record into contiguous groups of size k, averages observables within each group, and only then applies the linear leak and lag map to each group sequence. Candidate values of k receive distinct deterministic permutations. Because both averaging and the leak/lag operator are linear,

$$\frac{1}{G}\sum_{g}\mathcal{P}\left(\frac{1}{k}\sum_{n\in g} z_t^{(n)}\right) = \mathcal{P}\left(\frac{1}{N}\sum_{n=1}^{N} z_t^{(n)}\right), \qquad (6)$$

where $\mathcal{P}$ denotes the common leak-and-lag map. Thus every grouping has exactly the same mean feature sequence as EV; only the empirical within-group covariance differs.

The headline design averages over grouping seeds 17, 43, and 97. These are nested implementation repeats, not independent forecasting samples. Their role is to average over arbitrary group assignments before uncertainty is computed across trajectories or windows.

APPENDIX F
COMPLETED FOUR-TASK HARDWARE TRANSFER

Mackey–Glass, Santa Fe, Rössler, and sunspots use the same unmitigated ten-qubit circuit, local-Z readout, leak and ten-lag postprocessing, chronological development/test split, and tuning grids on `ibm_fez` and `ibm_marrakesh`. Each task–processor pair contributes one $T = 500$ acquisition with 400 ordered shots per step. The record is partitioned into twenty disjoint $N = 20$ blocks, each retaining the 10,000-execution budget used by the balanced simulation setting.

This hardware circuit is separate from the six-qubit expressive reservoir used in simulation. It uses one amplitude-encoded input qubit, nine Hadamard-initialized memory qubits, a depth-one ring of $R_{ZZ}(\pi/2)$ couplings, $R_X(\pi/4)$ rotations, and ten local-Z measurements. The hardware study tests the aggregation procedure on a second fixed reservoir realization. It is not a reproduction of the six-qubit feature map and does not compare qubit counts. The circuit and physical layout are fixed within each processor, so EV, Grouped, and Raw differ only in their classical use of the same ordered measurement record.

Santa Fe uses continuation window 5, $[4587, 5088)$, disjoint from the five simulation windows. It was acquired on July 12, 2026, under the same circuit, archived physical layouts, and analysis as the earlier hardware tasks. The Fez job `d99mf4t2su3c739kl9k0` used 55 QPU seconds; the Marrakesh job `d99mf52f47jc73a99fj0` used 56, for 111 seconds total. On Fez, Santa EV and Grouped NRMSE are 0.750 and 0.711; on Marrakesh they are 0.744 and 0.713. Grouped wins 19/20 blocks on each processor.

The paired intervals use the twenty disjoint shot blocks within each single acquisition. They are descriptive checks of shot-record organization and must not be interpreted as replication across calibration periods, jobs, or devices.

APPENDIX G
REPRODUCIBILITY COMMANDS AND ARTIFACT INVENTORY

All commands below run from the repository root. First regenerate the target-free diagnostic and check its serialized parameter hash:

```
D=scripts/\
run_expressive_reservoir_diagnostics.py
python "$D" --feature-grid-csv \
  results/expressive_reservoir/\
ideal_feature_grid.csv
```

Inspect the current four task job plan and verify the included evidence:

```
python scripts/run_robust_evaluation.py \
  --config configs/benchmark.json \
  --results-dir results/benchmark_dry_run \
  --dry-run
```

TABLE IV

BALANCED-SETTING MEAN TEST NRMSE FOR THE CLASSICAL AND IDEAL CONTROLS AND EV, GROUPED, RAW, AND COV-EV. BOLD MARKS THE BEST SAMPLED QUANTUM MEAN, NOT THE BEST VALUE ACROSS THE CLASSICAL AND IDEAL COLUMNS.

Task	Classical	Ideal QRC	EV	Grouped	Raw	Cov-EV
Mackey–Glass	0.098	0.190	0.484	0.439	0.586	**0.430**
Santa Fe laser	0.762	0.493	0.764	0.680	0.697	**0.677**
Rössler	0.096	0.054	0.414	0.379	0.580	**0.367**
Sunspots	0.387	0.474	0.553	0.517	0.558	**0.515**

TABLE V

BALANCED-SETTING PAIRED EFFECTS OVER INDEPENDENT TRAJECTORIES OR WINDOWS. VALUES ARE MEAN [95% CI] IN NRMSE.

Task	EV − Grouped	Grouped − Cov-EV
Mackey–Glass	0.0445 [0.0372, 0.0519]	0.0091 [0.0058, 0.0123]
Santa Fe laser	0.0848 [0.0174, 0.1522]	0.0026 [−0.0167, 0.0218]
Rössler	0.0349 [0.0139, 0.0559]	0.0116 [0.0021, 0.0212]
Sunspots	0.0360 [0.0103, 0.0616]	0.0021 [−0.0025, 0.0067]

```
python scripts/verify_santa_fe_manuscript.py
python scripts/plot_manuscript_figures.py
```

The completed IBM acquisitions cannot be reproduced without new QPU use. The released raw records can be reanalyzed and the hardware figure redrawn offline:

```
python scripts/analyze_qai_hardware.py
python scripts/analyze_qai_hardware.py \
  --subdir marrakesh
python scripts/plot_hardware_figure.py
```

The extension manifest preserves live availability, frozen input hashes, physical layouts, job identifiers, ordered-bitstring hashes, and charged time; credentials are neither stored nor released.

A. Artifact inventory

Frozen inputs: The current runnable protocol is `configs/benchmark.json`. The files `configs/qai_robust_protocol.json`, `configs/qai_robust_protocol_santa_fe.json`, and `results/expressive_reservoir/ideal_diagnostics.json` are retained because their hashes are recorded by the completed source manifests. **Simulation provenance:** the source manifests in `results/qai_robust_protocol_full/` and `results/qai_santa_fe_laser_screen/` record configuration, implementation, commit, data, and test-access hashes. The merge manifest in `results/qai_robust_protocol_santa_fe_manuscript/` records both source hashes and output provenance. Source `shards/` contain checkpoints. The merged `replicate_results.csv`, `cluster_summary.csv`, `cluster_paired_stats.csv`, and `partition_robustness_summary.csv` contain completed nested rows, cluster summaries, EV-paired differences, and grouping sensitivity. The paper directory contains the balanced table source and regenerated vector figures; the merged result directory contains the manuscript-facing tables and their hashes. **Hardware provenance:** `results/qai_hardware_santa_fe_extension/manifest.json` records the completed Santa acquisition and promotion into the two four-task archives under `results/qai_hardware/`. The raw shots, task inputs, layouts, operational manifests, and derived tables needed to audit the main-paper hardware figure are released together.

Build this standalone supplement with

```
cd paper
latexmk -pdf -interaction=nonstopmode \
  -halt-on-error supplement.tex
```

The bibliography and IEEE class are the same files used by the six-page main paper. The named artifact is available at github.com/eybmits/split-ensemble-qrc.

REFERENCES

[1] C. M. Bishop, "Training with noise is equivalent to Tikhonov regularization," *Neural Computation*, vol. 7, no. 1, pp. 108–116, 1995.

[2] A. Pérez-Salinas, A. Cervera-Lierta, E. Gil-Fuster, and J. I. Latorre, "Data Re-Uploading for a Universal Quantum Classifier," *Quantum*, vol. 4, p. 226, 2020.